

Reader's Workshop and Article Analysis

Topic: Software Evaluation for Qualitative Research

Preface

I was not surprised that Jim Pham and I were selected to report on this topic as our contribution to the Reader's Workshop. What was a bit surprising to both of us was that software applications to support qualitative research have been around for over 20 years. Software development in qualitative research has mirrored that in many other areas. Crude, clunky applications from the early days have been refined into powerful tools that have become essential requirements for the profession.

Jim and I divided the tasks of reporting and evaluation as follows: I would look at the general application of computer-assisted qualitative data analysis software (CAQDAS), and Jim would evaluate and demonstrate the latest version of NUD*IST, also known as NVivo 7, one of the leading application packages.

CAQDAS

The use of Computer-Assisted Qualitative Data Analysis Software (CAQDAS), also sometimes simply called Qualitative Data Analysis Software (QDAS or QDA software), has become increasingly popular, particularly in the field of social research. With this in mind, it is important to remind all that CAQDAS is simply a tool and is not a methodology. John Creswell, the author of our class text, describes the tool in this way: “The process used for QDA is the same for hand-coding or using a computer... the computer simply provides a means for storing data and easily accessing the codes.

Creswell goes on to list the following advantages of using computers as a tool to accomplish qualitative research:

- Provides an organized storage system.
- Can help researchers locate material quickly and easily.
- Encourage researchers to look closely at the data line by line.
- The concept mapping feature enables the researcher to visualize relationships by drawing a visual model.
- Allow a researcher to easily retrieve memos associated with codes, themes, or documents.

Creswell’s points were echoed in the online article [CAQDAS - A primer](#), from the Department of Social Sciences at Loughborough University: Computer-Assisted Qualitative Data Analysis Software (CAQDAS) searches, organizes, categorizes, and annotates textual and visual data. Programs of this type also frequently support theory-building by visualizing relationships among variables coded in the data. All CAQDAS applications support some form of text and code searches. Among the most common search functions are:

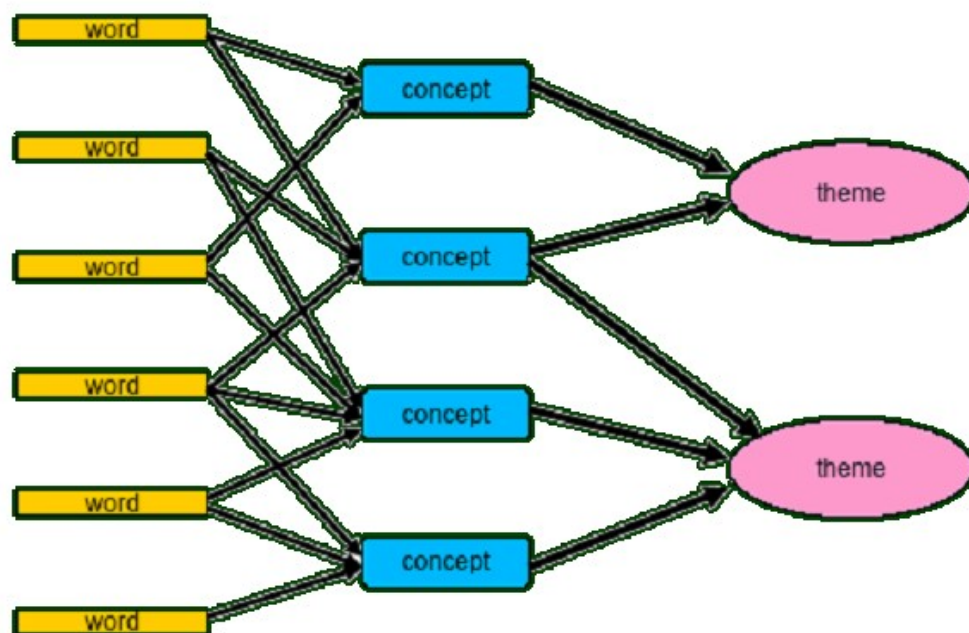
- Simple searches, which retrieve text strings from data and/or codes.
- Placeholder searches, which allow you to use placeholders for certain characters in your search.
- Boolean searches, which offer various combinations of searches using the Boolean operators AND, OR XOR, and NOT across various section of codes or texts.

- Proximity searches allow you to retrieve combinations of two or more text strings and/or codes, which occur in a definable proximity to each other.
- Fuzzy searches allow you to perform fuzzy searches that retrieve textual data. Such a type of search is particularly useful if you expect your data to contain typographic errors.

At the center of the value of a CAQDAS application is to provide the capability for the user to code data. A number of basic coding strategies:

- **Memos** small electronic Post-it™ notes, you can attach to all sorts of data bits.
- **Free coding** allows you to mark sections of data and attach a code to these sections.
- **In vivo coding** is a subtype of free coding, where a portion of text can be coded.
- **Automatic coding** of search results, or automatic recording of data according to previously specified queries, so-called "supercodes".
- **Software-generated coding algorithms suggest codes on the basis** of previously co-occurring codes.
- **Multimedia coding** allows you to code sequences of audio or video files and parts of pictures.

But some programs also allow you to link your codes in various ways and visualize these links in diagrams. Most new programs also have a number of ways to visualize co-occurrences of codes, to assist you in concept mapping and theory building.



Qualitative Research Methods and CAQDAS

Even though developers of CAQDAS would have you believe that their application supports any and all methodologies of qualitative research, some methodologies are much better supported by the software applications than others. I conclude my portion of this paper with a list of methodologies and how well each is supported by CAQDAS.

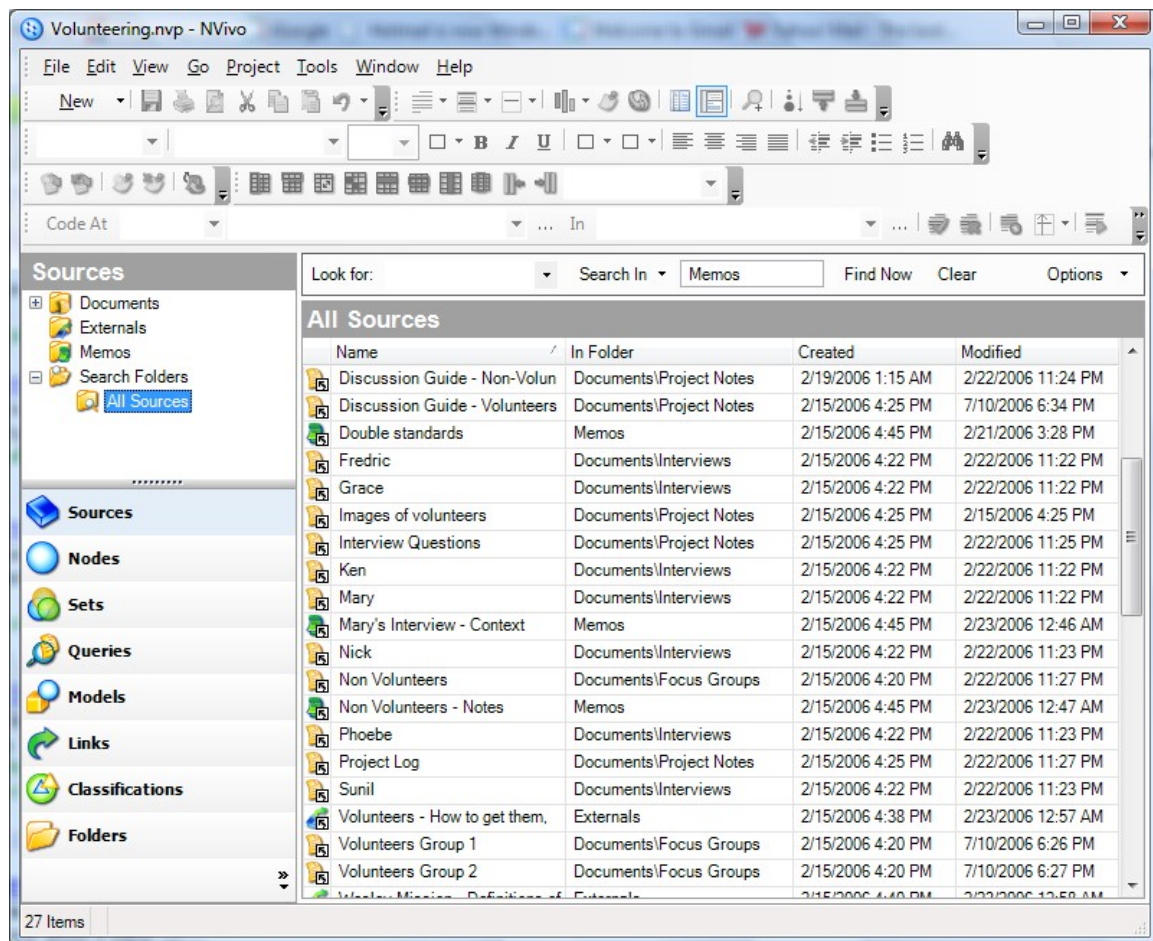
Methodology	Well Supported by CAQDAS	Enhanced by CAQDAS	Limited Potential
Grounded Theory	X		
Qualitative Content Analysis	X		
Ethnography	X		
Phenomenological	X		
Frame Analysis, in particular, for quantification purposes.		X	
Conversation Analysis		X	
Discourse Analysis			X
Visual Analysis			X
(Quantitative) Content Analysis			X
Comparative Analysis			X

Computer-Aided Qualitative Data Analysis Tools

NUD*IST (Non-numerical Unstructured Data Indexing Searching & Theorizing) is one of the many computer-aided qualitative data analysis tools from QSR International and it is a popular and powerful tool to support qualitative research. The latest version of NUD*IST is NVivo 7 and it has a trial version for 30 days from QSR International. American Evaluation association contains a lot of useful links to others software package for qualitative data analysis (<http://www.eval.org/Resources/QDA.htm>).

NUD.IST can help you:

1. “Manage” your data as you analyze documents such as interview transcripts, field notes, journal articles, papers, email archives or any other data you can save as a text file.
2. It will also facilitate your management of other data formats such as video and audio tapes.
3. You can search and code your data and automate many of the coding, searching, and reporting tasks.
4. You can write memos and add context-sensitive annotations to your data. As you progress through your analysis, you can use NUD.IST to document your analytic process and reflect on your role as a researcher.
5. There are many other features not listed here (See screenshot below).



References

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